

SCRIBE: Consistent Distributed Environment Modeling for Communication-Limited Robotic Exploration

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Abstract—Autonomous exploration of the ocean and other communication-limited environments requires agents to construct models from sparse measurements while using those models to determine where to sample next. Distributed teams can extend the scale and responsiveness of this process, but their ability to learn collectively depends on exchanging information across intermittent, bandwidth-constrained networks. Observation sharing grows with mission duration and sensor dimensionality, while conventional model-state fusion can become overconfident when model correlations are unknown. We introduce SCRIBE, a fixed-dimensional distributed environment modeling process for consistent model assimilation whenever communication is available. SCRIBE represents a scalar field through Gaussian basis functions and exploits the resulting linear information form to combine Information Kalman filtering, Covariance Intersection, and Metropolis-Hastings consensus. Agents can thereby exchange learned model states over any available peer link, without knowledge of the full communication graph or waiting for scheduled model-update windows. We evaluate SCRIBE with independent Vulcan information-based planners in twelve simulated four-robot trials under distance-limited communication and a complete blackout. The resulting trajectories achieve pooled posterior variance within 3.2% of ideal centralized sharing, indicating that consistent model assimilation can preserve effective distributed exploration through communication loss.

Index Terms—autonomous exploration, distributed estimation, informative path planning, intermittent communication, multi-robot systems.

I. INTRODUCTION

Autonomous exploration is both a learning and a decision problem. A robot entering an unfamiliar region must gather observations, construct a model of the environment, and use that model to determine which future observations will be most valuable. Statements of model confidence are central to this process: they provide a means of validating what has been learned and indicate where the explorer still lacks environmental understanding. Gaussian Processes and related uncertainty-aware models have consequently found broad use in adaptive sampling [1] and information-based path planning [2].

The need for this capability is particularly clear in the deep ocean. An autonomous underwater vehicle may be asked to localize a hydrothermal plume, characterize a cold seep, or follow a chemical gradient using only the sparse measurements gathered during its mission; autonomous cold-seep exploration is one demonstrated example [3]. A team can observe a larger region and pursue several scientific targets concurrently, but underwater acoustic communication is intermittent, bandwidth-limited, and strongly dependent on vehicle separation [1]. Each robot must therefore continue

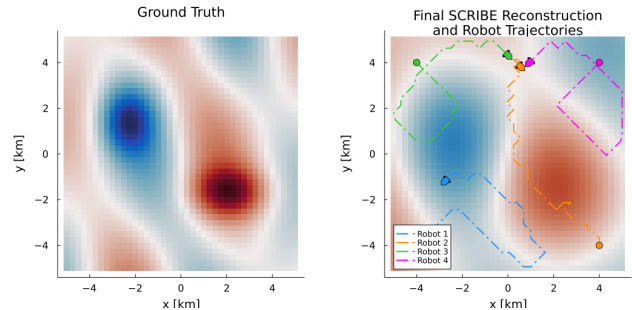


Fig. 1. Simulated scalar field over a 10 km $\times$ 10 km domain (left) and one robot's final SCRIBE posterior mean (right). Large circles mark the robots' initial locations; dash-dot paths terminate in oriented robot symbols. The displayed model is the online state at the final sampling round, rather than a reconstruction formed afterward by replaying the observation history.

learning and planning without contact, and incorporate peers' learned model states whenever communication resumes.

Distributing model learning, however, is a non-trivial process, particularly when the agents' estimates must remain consistent with each other. Existing approaches share gathered or synthetic observations and relearn from the collective data [4], [5], or distribute Gaussian-process model updates across agents [6]. Observation sharing becomes computationally and communicatively burdensome as the history grows, especially when cameras, sonars, and spectrometers produce high-dimensional records. Sharing the learned model state is compact, but states that have interacted are correlated; treating them as independent evidence repeatedly counts the same observations and overstates certainty [7]. That failure directly affects exploration, since an overconfident model makes an unobserved region appear less informative than it is.

Finite-dimensional approximations reduce the cost of distributed Gaussian regression [8]. Related systems couple distributed models to informative planning [9], [10] or target search [11]. These results motivate compact model learning whose confidence remains consistent as communication changes.

We address this problem with SCRIBE: Signal-aware Collaboration for Reliable Information-based Estimation. SCRIBE combines a fixed-dimensional approximation of a Gaussian Process with a linear update that remains consistent under intermittent communication. Its principal contributions are threefold. First, an Information Kalman Filter (IKF) updates the coefficients of a Gaussian scalar-field (GSF) model, making observation information additive and distributable. Second, Covariance Intersection (CI) reconciles potentially divergent priors without requiring their unknown cross-

correlations [7], [12]. Third, Metropolis–Hastings Markov Chain (MHMC) weights distribute new information using only the identities and degrees of presently available neighbors, rather than complete knowledge of the communication graph [13]. SCRIBE brings these elements together following the hybrid consensus–CI construction for unreliable networks [14]. Agents can consequently exchange learned model states whenever a peer link becomes available, without a full-graph view or a scheduled team-wide update.

II. DISTRIBUTED ENVIRONMENT MODELING

A. Fixed-Dimensional Information State

SCRIBE represents a scalar field through n_ϕ shared GSF bases [8], [15],

$$\mathcal{F}(x, k) = \psi^T(x) \phi(k), \quad \psi_i(x) = \tau_i^{-1} e^{-\|x - \mu_i\|^2 / \sigma_i^2}. \quad (1)$$

Only the coefficient vector ϕ is estimated online. For $z_j = H_j \phi + v_j$, agent j maintains information matrix $Y_j = P_j^{-1}$ and vector $y_j = Y_j \hat{\phi}_j$. A measurement contributes

$$\delta I_j = H_j^T R_j^{-1} H_j, \quad \delta i_j = H_j^T R_j^{-1} z_j. \quad (2)$$

The exchanged learned model state comprises fixed-dimensional information vectors and matrices determined by n_ϕ . Its size is independent of mission duration; replaying k observations grows as $O(kd_z)$. Raw sensor products can be processed locally before entering the shared field model.

Algorithm 1 Distributed Scalar Field Modeling of Environment at Agent j

Require: GSF parameters $\{\mu_i, \sigma_i, \tau_i\}_{i=1}^{n_\phi}$
1: initialize (y_j, Y_j) and $\hat{\mathcal{F}}_j(x, 0) = \psi^T(x) \hat{\phi}_j(0)$
2: **while** executing sampling mission **do**
3: $(z_j, X_j) \leftarrow \text{GATHERSAMPLES}$
4: $(y_j, Y_j) \leftarrow \text{DISTRIBUTEDUPDATE}(\psi, y_j, Y_j, z_j, X_j)$
5: $\hat{\phi}_j \leftarrow Y_j^{-1} y_j$
6: $\hat{\mathcal{F}}_j(x) \leftarrow \psi^T(x) \hat{\phi}_j$
7: **end while**

B. Fusion under an Unstable Network

Current innovations are new evidence; priors can contain an unknown amount of shared history. SCRIBE deliberately treats them differently. CI chooses simplex weights that minimize the covariance of the fused neighbor priors without assuming their cross-correlations. In parallel, innovations are averaged with $\gamma_{jm} = [1 + \max(|\mathcal{N}_j|, |\mathcal{N}_m|)]^{-1}$. These MHMC weights require only current neighbor degrees, rather than knowledge of the global or future graph. After local convergence over a reachable component of size n_c ,

$$(y_j, Y_j) = (\tilde{y}_j, \tilde{Y}_j) + n_c(\bar{\delta i}_j, \bar{\delta I}_j). \quad (3)$$

CI prevents prior agreement from masquerading as a new observation; MHMC propagates genuinely new information. This separation permits repeated, peer-level fusion when links become available, rather than only during scheduled team-wide update windows.

Algorithm 2 SCRIBE: Distributed Update for Agent j under unstable networks

Require: Model ψ ; state (y_j, Y_j) ; sample (z_j, X_j)
1: $(y_j^-, Y_j^-) \leftarrow \text{LOCALPRIORUPDATE}(y_j, Y_j)$
2: $H_j \leftarrow \psi(X_j)$
3: $\delta i_j \leftarrow H_j^T R_j^{-1} z_j$; $\delta I_j \leftarrow H_j^T R_j^{-1} H_j$
4: initialize prior and innovation consensus variables
5: **while** local fusion has not converged **do**
6: exchange learned model-state components and degree with neighbors
7: fuse neighbor priors using CI
8: average neighbor innovations using MHMC
9: **end while**
10: **return** $(\tilde{y}_j, \tilde{Y}_j) + n_c(\bar{\delta i}_j, \bar{\delta I}_j)$

III. EXPERIMENTAL EVIDENCE

We first evaluate SCRIBE in simulation, independently of planning, using four agents on fixed sampling paths. A continuous-sharing check reproduces the centralized IKF to numerical precision. We then restrict every communication opportunity to disjoint, distance-limited peer pairs, remove all links for the middle third of the experiment, and restore only the same limited-range communication. Across 30 simulated trials with a truth represented by the GSF model, SCRIBE reaches a final RMSE of 0.1170 ± 0.0114 , compared with 0.1007 ± 0.0121 for centralized sharing, 0.1175 ± 0.0115 for CI alone, 0.2881 ± 0.0204 for the IKF without CI, and 0.8136 ± 0.0175 without communication. CI prevents the divergent priors formed during the blackout from being treated as independent, while the IKF innovation update recovers information that CI alone can only conservatively intersect.

To demonstrate SCRIBE’s effectiveness for exploration, we use four robots to sample a simulated field for 36 rounds under the same limited communication and blackout schedule. Five comparisons isolate the sharing mechanism. *KF only* sums current innovations without reconciling divergent priors; *CI only* intersects local posteriors without summing independent innovations; and *last-two observations* sends the two newest unseen records per available link. *No communication* retains local data, while *centralized sharing* provides every observation as an ideal bound. Every distributed method obeys the range limit: the largest component has two robots, and the blackout has no links.

We use Vulcan [16], here in a single-agent implementation for simplicity. Each robot plans from its own current model; there is no joint allocation or collaborative tree search. This isolates the information-reward structure supplied to otherwise decoupled planners. Figure 1 shows the online reconstruction and paths. In Fig. 2, SCRIBE’s final per-agent RMSE is 0.523 ± 0.060 , compared with 0.458 ± 0.052 centrally, 0.564 ± 0.084 for CI only, 0.730 ± 0.081 for KF only, 0.653 ± 0.067 for last-two sharing, and 1.041 ± 0.053 without communication.

To evaluate exploration separately from fusion error, we pool the samples collected under each regime and construct one evaluation model. The right panel of Fig. 2 reports its RMSE against the ground truth; differences therefore arise from where the decoupled planners chose to sample,

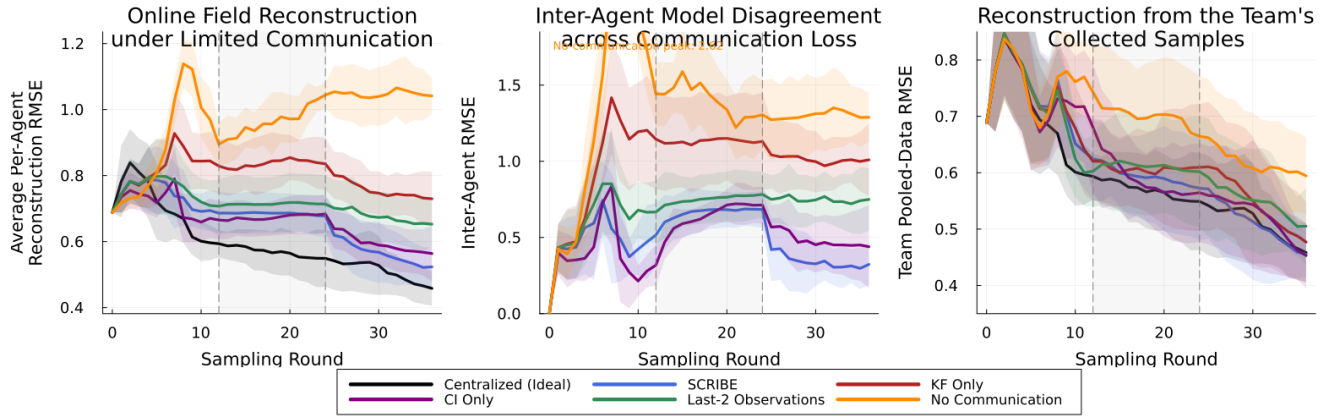


Fig. 2. Twelve simulated four-robot trials of 36 rounds. Communication is distance- and bandwidth-limited to one in-range peer per robot in rounds 1–12 and 25–36; the shaded interval is a complete blackout. Curves show the mean and one standard deviation. The right panel centrally pools only the samples collected under each regime, isolating trajectory quality from fusion error. The middle axis is truncated to resolve the communicating methods; the annotation gives the no-communication peak. Lower values are better.

rather than from how the agents fused their models. SCRIBE reaches a final pooled-data RMSE of 0.455, compared with 0.458 for centralized sharing and 0.594 without communication. We also measure the integrated posterior variance used to define the information reward on a common spatial grid. SCRIBE reaches 0.0488, within 3.2% of the centralized 0.0473; CI only reaches 0.0528, KF only 0.0617, last-two sharing 0.0651, and no communication 0.1074. The trajectories selected using SCRIBE therefore cover nearly as much information as the ideal-sharing trajectories, even though no distributed planner ever has access to a team-wide communication component.

SCRIBE communicates a learned model state whose dimension is determined by n_ϕ , rather than by mission duration or raw-sensor dimension. Although the simulated records are scalar, high-dimensional sonar, camera, or spectrometer products can be processed locally before entering the learned field model. Direct observation sharing must retain and transmit those records, a distinction that grows with the mission.

IV. CONCLUSION

SCRIBE provides a compact and consistent environment model for distributed exploration under limited communication. It remains locally useful through a communication blackout and can incorporate learned model states whenever peer links become available afterward. Although motivated by deep-ocean sampling, the same requirement arises in planetary, subterranean, and other remote missions where exploration cannot depend on continuous connectivity. The model update is independent of the planning architecture, and can consequently support coordinated planners such as Multi-Agent Vulcan [17] in future work.

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